

Backup Types — Detailed Lecture Notes

JKSSB Computer Awareness

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11.1 Backup, storage and archive

These three words describe different jobs, and exam questions often test the distinction. **Storage** is where your working data lives day to day — the primary copy you actually open and edit. A **backup** is a separate copy kept so that data can be recovered after loss, damage, corruption or an attack. An **archive** is older data retained for the long term, usually moved out of routine use. A backup exists for recovery; an archive exists for retention. Neither is simply “another folder” on the same disk.

Definition (Backup). *A separate copy of data kept for the purpose of recovery after loss, damage, corruption or attack.*

Definition (Archive). *Older data kept for long-term retention, normally not used in everyday work.*

The office desktop of the running example (KB EX-01) has one primary copy of its documents. If the only copy is the one on the desktop’s own drive, then the user has storage but no backup. The moment a second, independent copy exists, recovery becomes possible.

11.2 Full, incremental and differential backups

A backup type decides *how much* is copied on each run. The three basic types differ only in the answer to one question: since which earlier point are changes included?

Definition (Full backup). *Copies every selected file and folder (or the entire volume) on every run. It is the recovery baseline for the other types.*

Definition (Incremental backup). *Copies only the files changed or added since the last backup of any type — whether that was a full, an incremental or a differential backup.*

Definition (Differential backup). *Copies everything changed or added since the last full backup, not since the previous backup.*

The single phrase that separates incremental from differential is therefore the word *full*. An incremental looks only as far back as the immediately preceding backup; a differential always looks back to the last full backup, so it keeps growing until the next full backup resets it.

Type	What it copies	Restore requirement
Full	Everything selected, every run.	One complete set. Fastest single restore, but largest and slowest to create.
Incremental	Changes since the last backup of any type.	The last full backup plus every incremental backup in sequence.
Differential	Changes since the last full backup.	The last full backup plus only the most recent differential.

There is a consistent trade-off. Smaller and faster backups produce longer or more complex restores. A full backup is heavy to create but trivial to restore; an incremental is light to create but may need many files chained together at restore time; a differential sits in between.

Example 11.2.1. *On Monday the office desktop is given a full backup. On Tuesday, a single document changes. With an incremental strategy, Tuesday's backup contains only that one change. With a differential strategy, Tuesday's backup contains that change measured against Monday's full backup. If only that one file changed, the two results look identical on Tuesday — the difference appears on Wednesday, when the incremental again holds only Wednesday's change, while the differential holds Tuesday's change plus Wednesday's change.*

11.3 Restore scenarios

Restore behaviour is the part candidates most often miss, because it is not visible on the day of the backup — only when recovery is needed.

Example 11.3.1 (Restore from an incremental chain). *A full backup is taken Sunday. Incrementals are taken Monday, Tuesday and Wednesday. On Thursday the disk fails. To recover Wednesday's data you need the Sunday full backup and then all three incrementals — Monday, Tuesday and Wednesday — applied in order. Missing any one link loses the changes it carried.*

Example 11.3.2 (Restore from a differential chain). *A full backup is taken Sunday. Differentials are taken Monday, Tuesday and Wednesday. On Thursday the disk fails. To recover Wednesday's data you need only the Sunday full backup and the most recent differential — Wednesday's. The earlier differentials are not required, because Wednesday's already includes all changes since Sunday.*

Both scenarios produce the same recovered data. The difference is operational: the incremental restore is longer and depends on every tape or file in the chain; the differential restore is shorter and depends on only two sets.

11.4 Image and file-level backups

Beyond the three time-based types, two coverage-based terms appear in exams.

Definition (Image backup (system image)). *A complete clone of a system — operating system, settings, applications and data — that can restore a whole machine.*

Definition (File-level backup). *A backup of selected files and folders rather than a whole system.*

An image restores a working machine in one step, which makes it ideal after a drive failure or a corrupted operating system. A file-level backup is smaller and more flexible for everyday

documents, but recovering a machine from it means reinstalling the operating system first and then restoring the files. The office desktop could use an image for disaster recovery and file-level backups for its working documents.

11.5 The 3-2-1 rule, off-site and offline copies

The **3-2-1 rule** is the standard exam reminder for resilient backup:

1. **3** copies of the data — the original plus two backups.
2. On **2** different types of media (for example, an external disk and cloud storage).
3. With **1** copy kept **off-site**, at a different location.

The numbers map to copies, media and location, in that order. A common trap is to swap them, or to read the “2” as two off-site copies.

An **off-site** copy is kept at a different physical place, so a single fire, flood or theft cannot destroy every copy at once. An **offline** copy is disconnected from the network; it is especially useful against ransomware, which spreads through connected drives. A **cloud backup** is an off-site copy held on remote servers. These are not mutually exclusive: the office desktop’s cloud backup is both off-site and, when not mounted, can serve as an offline copy as well.

Example 11.5.1. *The office desktop keeps its working files on the local SSD (copy 1), an external USB disk (copy 2, different medium, kept offline), and a cloud account (copy 3, off-site). This satisfies the 3-2-1 rule: three copies, two media types, one off-site.*

11.6 Why RAID is not a backup

RAID (**Redundant Array of Independent Disks**) combines several disks for performance or redundancy. If one disk fails, a mirrored or parity RAID keeps the system running. That protects *uptime*, not *recovery*. RAID is not a backup because it does not protect against the things a backup exists for: accidental deletion, corruption or ransomware. When a file is deleted, or an encrypted copy is written by ransomware, the change is written to every member of the RAID array at once. There is no earlier clean copy to go back to. Redundancy duplicates the current state — including the damage.

11.7 Backup rotation

Backup rotation means keeping several dated backup sets and reusing them on a fixed schedule, commonly the grandfather–father–son scheme (daily, weekly, monthly). Rotation matters because a problem such as silent corruption may not be noticed for days; if the single backup set has already been overwritten with the corrupted data, there is no clean copy left. Keeping multiple versions across time ensures that an older, clean copy remains available.

Exercises

1. Define full, incremental and differential backup in one line each.
2. A full backup runs Sunday; differentials run Monday to Wednesday. The disk fails on Thursday. Which sets are needed to restore Wednesday’s data?

3. State the 3-2-1 rule and explain what each number refers to.
4. Why is RAID not a substitute for a backup? Give one concrete danger.
5. Explain the purpose of backup rotation.

Solutions

1. Full: copies everything selected each run. Incremental: copies changes since the last backup of any type. Differential: copies changes since the last full backup.
2. The Sunday full backup and the most recent (Wednesday) differential only. The earlier differentials are unnecessary because Wednesday's includes all changes since the full backup.
3. Three copies of the data, on two different types of media, with one copy kept off-site.
4. RAID provides redundancy for uptime, not recovery. A deletion, corruption or ransomware write is applied to every RAID member at once, so no earlier clean copy survives.
5. Rotation retains several dated backup versions so that an older, clean copy is still available if corruption is discovered only after the newest backup has already been overwritten.